IB Economics - Topic 2.6: Teacher and AI Grading Guide

Elasticity of Supply | Questions 1-50

Scope and assessment status

Each original paired question has (a) 10 marks and (b) 15 marks. Questions 1-45 are SL/HL. Questions 46-50 assess the HL-only comparison of primary commodities and manufactured products. Difficulty: 10 Easy, 25 Medium and 15 Hard pairs, distributed through the bank. Level and difficulty are independent.

Current IB Topic 2.6 covers PES formula/degrees, time, factor mobility, unused capacity, storage, and rate at which costs rise; diagrams and PES or unknown price/quantity calculations. HL only: reasons why PES of primary commodities is generally lower than PES of manufactured goods.

The uploaded older IB textbook Chapter 3 section 3.4 supports core PES concepts. Its examples and numbering are supplementary; current IB syllabus sets scope.

These are original adapted Paper 1-style practice questions, not official IB examination or marking material. The question bank revisits syllabus mechanisms through distinct explanatory and evaluative tasks; the pairs are alternatives for practice, not separate official learning outcomes.

Coverage blueprint

PES basics and degrees: 1-10

Time and factors: 11-20

Capacity and storage: 21-30

Rising costs and interacting determinants: 31-40

Estimation: 41-45

HL primary/manufactured comparison: 46-50

Marking principles

Use best-fit assessment for each part, not a fixed marks-per-keyword checklist. Credit accurate alternatives addressing the task. Do not require every example or qualification below.

Part (a): 10 marks

Practice bands: 0; 1-2 limited; 3-4 mainly descriptive; 5-6 partial explanation; 7-8 clear and accurate causal analysis; 9-10 fully developed explanation with any required diagram or calculation. These abbreviated bands are not official IB descriptors.

Part (b): 15 marks

Practice bands: 0; 1-3 very limited; 4-6 superficial; 7-9 developed but weakly balanced; 10-12 balanced analysis and evaluation; 13-15 sustained, applied evaluation and supported judgment. A named case alone is not developed application.

Diagram and AI grading

Check P vertical and Q supplied horizontal, correct response for a specified price change, matched axes when comparing relative elasticity, and PES = % change Qs / % change P. Vertical S has zero PES; horizontal S has infinite PES. Do not confuse a shift with a movement.

Record separate marks for (a) and (b), band evidence, diagram or numerical features if relevant, economic misconceptions, quality of real-world application, and one actionable improvement per part. If a student drawing is unclear, flag the uncertainty rather than assuming correctness.

Question 1

Easy | SL/HL | 25 marks

Topic 2.6: PES basics

(a) [10] Explain what price elasticity of supply measures and how it differs from the slope of a supply curve.

(b) [15] Using real-world examples, discuss whether an upward-sloping supply curve reveals how responsive producers are to price.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. PES compares percentage quantity supplied change with percentage own-price change.

Expected diagram or calculation. P vertical and Q supplied horizontal when drawn; calculation percentage Qs change divided by percentage P change.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes pes basics from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. PES compares percentage quantity supplied change with percentage own-price change.

Counterargument and conditions. A point/interval estimate may vary with conditions and is not a supply shift.

Application and real-world example. USDA crop production and prices are context; do not claim causal PES from two observations.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 2

Medium | SL/HL | 25 marks

Topic 2.6: PES basics

(a) [10] Price rises by 10% and quantity supplied rises by 20%. Calculate PES and interpret the result.

(b) [15] Using real-world examples, evaluate whether this estimate predicts the response to a further price rise.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. PES=2, elastic for the stated interval; conditions can vary.

Expected diagram or calculation. P vertical and Q supplied horizontal when drawn; calculation percentage Qs change divided by percentage P change.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes pes basics from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. PES=2, elastic for the stated interval; conditions can vary.

Counterargument and conditions. A point/interval estimate may vary with conditions and is not a supply shift.

Application and real-world example. USDA crop production and prices are context; do not claim causal PES from two observations.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 3

Hard | SL/HL | 25 marks

Topic 2.6: PES basics

(a) [10] A good has PES 0.4. Explain the expected effect of a 5% price rise, assuming PES remains constant.

(b) [15] Using real-world examples, discuss whether low PES implies firms never increase output.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. Quantity supplied rises about 2%; inelastic is not zero.

Expected diagram or calculation. P vertical and Q supplied horizontal when drawn; calculation percentage Qs change divided by percentage P change.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes pes basics from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Quantity supplied rises about 2%; inelastic is not zero.

Counterargument and conditions. A point/interval estimate may vary with conditions and is not a supply shift.

Application and real-world example. USDA crop production and prices are context; do not claim causal PES from two observations.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 4

Medium | SL/HL | 25 marks

Topic 2.6: PES basics

(a) [10] Explain why PES is usually non-negative for an upward-sloping supply curve.

(b) [15] Using real-world examples, evaluate whether the law of supply guarantees an elastic supply response.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. Price and quantity supplied move together, but proportional response may be small.

Expected diagram or calculation. P vertical and Q supplied horizontal when drawn; calculation percentage Qs change divided by percentage P change.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes pes basics from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Price and quantity supplied move together, but proportional response may be small.

Counterargument and conditions. A point/interval estimate may vary with conditions and is not a supply shift.

Application and real-world example. USDA crop production and prices are context; do not claim causal PES from two observations.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 5

Medium | SL/HL | 25 marks

Topic 2.6: PES basics

(a) [10] Explain why the same absolute change in output can represent different PES values for two firms.

(b) [15] Using real-world examples, discuss whether raw output increases are sufficient to compare supply responsiveness.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. Different initial output and percentage price changes produce different ratios.

Expected diagram or calculation. P vertical and Q supplied horizontal when drawn; calculation percentage Qs change divided by percentage P change.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes pes basics from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Different initial output and percentage price changes produce different ratios.

Counterargument and conditions. A point/interval estimate may vary with conditions and is not a supply shift.

Application and real-world example. USDA crop production and prices are context; do not claim causal PES from two observations.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 6

Easy | SL/HL | 25 marks

Topic 2.6: PES degrees

(a) [10] Using appropriate diagrams, explain the difference between relatively elastic and relatively inelastic supply.

(b) [15] Using real-world examples, evaluate whether supply curves' visual steepness alone identifies PES.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. Compare Q percent response to P percent change at specified scales and points.

Expected diagram or calculation. P vertical, Q horizontal; compare relatively flat/steep S and show horizontal or vertical extreme where relevant.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes pes degrees from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Compare Q percent response to P percent change at specified scales and points.

Counterargument and conditions. Horizontal/vertical are limiting cases; axis scaling and relevant point matter.

Application and real-world example. Original art compared with agricultural production.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 7

Hard | SL/HL | 25 marks

Topic 2.6: PES degrees

(a) [10] Explain the meaning of unitary PES and calculate quantity response to a 6% price rise when PES equals one.

(b) [15] Using real-world examples, discuss whether unitary PES is likely at all output levels.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. Quantity supplied rises 6% under stated conditions; the value can vary along curves.

Expected diagram or calculation. P vertical, Q horizontal; compare relatively flat/steep S and show horizontal or vertical extreme where relevant.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes pes degrees from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Quantity supplied rises 6% under stated conditions; the value can vary along curves.

Counterargument and conditions. Horizontal/vertical are limiting cases; axis scaling and relevant point matter.

Application and real-world example. Original art compared with agricultural production.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 8

Medium | SL/HL | 25 marks

Topic 2.6: PES degrees

(a) [10] Using diagrams, explain the meanings of perfectly elastic and perfectly inelastic supply.

(b) [15] Using real-world examples, evaluate whether either extreme describes most production processes.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. Horizontal S infinite responsiveness at stated P, vertical S zero Q response.

Expected diagram or calculation. P vertical, Q horizontal; compare relatively flat/steep S and show horizontal or vertical extreme where relevant.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes pes degrees from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Horizontal S infinite responsiveness at stated P, vertical S zero Q response.

Counterargument and conditions. Horizontal/vertical are limiting cases; axis scaling and relevant point matter.

Application and real-world example. Original art compared with agricultural production.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 9

Hard | SL/HL | 25 marks

Topic 2.6: PES degrees

(a) [10] Explain why a fixed stock of original paintings might have PES of zero.

(b) [15] Using real-world examples, discuss whether every good with a limited current stock is perfectly inelastic in the long run.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. Original fixed units cannot be produced; reproductions are different goods.

Expected diagram or calculation. P vertical, Q horizontal; compare relatively flat/steep S and show horizontal or vertical extreme where relevant.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes pes degrees from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Original fixed units cannot be produced; reproductions are different goods.

Counterargument and conditions. Horizontal/vertical are limiting cases; axis scaling and relevant point matter.

Application and real-world example. Original art compared with agricultural production.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 10

Medium | SL/HL | 25 marks

Topic 2.6: PES degrees

(a) [10] Price falls by 8% and quantity supplied falls by 4%. Calculate PES and explain whether supply is elastic.

(b) [15] Using real-world examples, evaluate whether such a result would remain stable as production conditions change.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. PES=0.5, inelastic over the interval.

Expected diagram or calculation. P vertical, Q horizontal; compare relatively flat/steep S and show horizontal or vertical extreme where relevant.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes pes degrees from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. PES=0.5, inelastic over the interval.

Counterargument and conditions. Horizontal/vertical are limiting cases; axis scaling and relevant point matter.

Application and real-world example. Original art compared with agricultural production.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 11

Easy | SL/HL | 25 marks

Topic 2.6: Time

(a) [10] Explain why supply may be less price elastic immediately after a price rise than a year later.

(b) [15] Using real-world examples, discuss whether time alone guarantees a more elastic response.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. Adjusting labour, equipment and production plans takes time.

Expected diagram or calculation. Use P/Q axes and short-run versus long-run S only if requested, with consistent scale.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes time from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Adjusting labour, equipment and production plans takes time.

Counterargument and conditions. Adjustment horizon alone does not ensure resources, demand and capacity are available.

Application and real-world example. USDA crop planting intentions and citrus seasons.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 12

Medium | SL/HL | 25 marks

Topic 2.6: Time

(a) [10] Explain how a farming season limits the short-run PES of a crop.

(b) [15] Using real-world examples, evaluate whether agricultural supply must remain unresponsive in future seasons.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. Planting and harvesting cannot be completed instantly, but next season acreage can change.

Expected diagram or calculation. Use P/Q axes and short-run versus long-run S only if requested, with consistent scale.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes time from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Planting and harvesting cannot be completed instantly, but next season acreage can change.

Counterargument and conditions. Adjustment horizon alone does not ensure resources, demand and capacity are available.

Application and real-world example. USDA crop planting intentions and citrus seasons.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 13

Hard | SL/HL | 25 marks

Topic 2.6: Time

(a) [10] Using appropriate diagrams, explain how the responsiveness of supply can differ between the short run and the long run.

(b) [15] Using real-world examples, discuss whether the same supply curve can be used for both periods.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. Different feasible adjustments make longer-run S relatively more responsive at comparable axes.

Expected diagram or calculation. Use P/Q axes and short-run versus long-run S only if requested, with consistent scale.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes time from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Different feasible adjustments make longer-run S relatively more responsive at comparable axes.

Counterargument and conditions. Adjustment horizon alone does not ensure resources, demand and capacity are available.

Application and real-world example. USDA crop planting intentions and citrus seasons.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 14

Medium | SL/HL | 25 marks

Topic 2.6: Time

(a) [10] Explain why a firm with a long construction period may not expand output immediately after a price increase.

(b) [15] Using real-world examples, evaluate the importance of production time for PES compared with other determinants.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. Capacity cannot be ready instantly; unused capacity may provide interim response.

Expected diagram or calculation. Use P/Q axes and short-run versus long-run S only if requested, with consistent scale.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes time from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Capacity cannot be ready instantly; unused capacity may provide interim response.

Counterargument and conditions. Adjustment horizon alone does not ensure resources, demand and capacity are available.

Application and real-world example. USDA crop planting intentions and citrus seasons.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 15

Medium | SL/HL | 25 marks

Topic 2.6: Time

(a) [10] Explain why a temporary price rise might elicit a smaller investment response than a lasting one.

(b) [15] Using real-world examples, discuss whether future price expectations matter when estimating PES.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. Firms may avoid costly expansion for a short-lived signal.

Expected diagram or calculation. Use P/Q axes and short-run versus long-run S only if requested, with consistent scale.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes time from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Firms may avoid costly expansion for a short-lived signal.

Counterargument and conditions. Adjustment horizon alone does not ensure resources, demand and capacity are available.

Application and real-world example. USDA crop planting intentions and citrus seasons.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 16

Easy | SL/HL | 25 marks

Topic 2.6: Factor mobility

(a) [10] Explain how mobility of factors of production affects PES.

(b) [15] Using real-world examples, evaluate whether a high price is enough to draw resources into an industry.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. Transferable labour and machinery permit quicker output adjustment.

Expected diagram or calculation. P/Q axes if drawn; explain percentage responsiveness for equal percentage price change.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes factor mobility from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Transferable labour and machinery permit quicker output adjustment.

Counterargument and conditions. Capital and skills may remain specialised despite higher price.

Application and real-world example. USDA crop acreage responses conditional on land suitability.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 17

Hard | SL/HL | 25 marks

Topic 2.6: Factor mobility

(a) [10] Explain why specialised equipment may make supply relatively price inelastic.

(b) [15] Using real-world examples, discuss whether training alone can remove constraints on supply.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. Equipment cannot easily be redeployed or obtained; workers are only one input.

Expected diagram or calculation. P/Q axes if drawn; explain percentage responsiveness for equal percentage price change.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes factor mobility from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Equipment cannot easily be redeployed or obtained; workers are only one input.

Counterargument and conditions. Capital and skills may remain specialised despite higher price.

Application and real-world example. USDA crop acreage responses conditional on land suitability.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 18

Medium | SL/HL | 25 marks

Topic 2.6: Factor mobility

(a) [10] Using appropriate diagrams, explain how supply responsiveness may differ between an industry using transferable labour and one using specialised skills.

(b) [15] Using real-world examples, evaluate how important labour mobility is to PES.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. Compare quantity response at the same percentage price change with consistent axes.

Expected diagram or calculation. P/Q axes if drawn; explain percentage responsiveness for equal percentage price change.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes factor mobility from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Compare quantity response at the same percentage price change with consistent axes.

Counterargument and conditions. Capital and skills may remain specialised despite higher price.

Application and real-world example. USDA crop acreage responses conditional on land suitability.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 19

Hard | SL/HL | 25 marks

Topic 2.6: Factor mobility

(a) [10] Explain why geographical immobility of productive resources can limit a firm's response to a local price rise.

(b) [15] Using real-world examples, discuss whether producers in one region can always respond to higher prices in another.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. Location and transport prevent immediate reallocation to local production.

Expected diagram or calculation. P/Q axes if drawn; explain percentage responsiveness for equal percentage price change.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes factor mobility from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Location and transport prevent immediate reallocation to local production.

Counterargument and conditions. Capital and skills may remain specialised despite higher price.

Application and real-world example. USDA crop acreage responses conditional on land suitability.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 20

Medium | SL/HL | 25 marks

Topic 2.6: Factor mobility

(a) [10] Explain how readily available rented machinery may affect a producer's PES.

(b) [15] Using real-world examples, evaluate whether factor mobility guarantees elastic supply.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. Access can speed expansion, but storage, capacity and costs may still constrain.

Expected diagram or calculation. P/Q axes if drawn; explain percentage responsiveness for equal percentage price change.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes factor mobility from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Access can speed expansion, but storage, capacity and costs may still constrain.

Counterargument and conditions. Capital and skills may remain specialised despite higher price.

Application and real-world example. USDA crop acreage responses conditional on land suitability.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 21

Easy | SL/HL | 25 marks

Topic 2.6: Spare capacity

(a) [10] Explain why unused productive capacity tends to make supply more price elastic.

(b) [15] Using real-world examples, discuss whether spare equipment alone is sufficient to raise output.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. Idle machines can be activated without building new plant if complementary inputs available.

Expected diagram or calculation. P/Q axes; compare Q response to the same P change at matched scales.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes spare capacity from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Idle machines can be activated without building new plant if complementary inputs available.

Counterargument and conditions. Idle capacity requires complementary workers and inputs.

Application and real-world example. Hotel rooms and factories as named real markets where documented.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 22

Medium | SL/HL | 25 marks

Topic 2.6: Spare capacity

(a) [10] Explain why a firm operating at full capacity may respond little to a price increase.

(b) [15] Using real-world examples, evaluate whether a capacity limit makes long-run supply inelastic.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. Near-term expansion needs investment; later capacity can increase.

Expected diagram or calculation. P/Q axes; compare Q response to the same P change at matched scales.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes spare capacity from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Near-term expansion needs investment; later capacity can increase.

Counterargument and conditions. Idle capacity requires complementary workers and inputs.

Application and real-world example. Hotel rooms and factories as named real markets where documented.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 23

Hard | SL/HL | 25 marks

Topic 2.6: Spare capacity

(a) [10] Using diagrams, explain how different levels of unused capacity can alter the supply response to the same price rise.

(b) [15] Using real-world examples, discuss whether a steeper short-run supply curve proves a firm is operating at capacity.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. Show larger/smaller Q movement for specified P change, keeping market and axes comparable.

Expected diagram or calculation. P/Q axes; compare Q response to the same P change at matched scales.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes spare capacity from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Show larger/smaller Q movement for specified P change, keeping market and axes comparable.

Counterargument and conditions. Idle capacity requires complementary workers and inputs.

Application and real-world example. Hotel rooms and factories as named real markets where documented.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 24

Medium | SL/HL | 25 marks

Topic 2.6: Spare capacity

(a) [10] Explain why a vacant hotel room may permit a quicker supply response than building a new hotel.

(b) [15] Using real-world examples, evaluate whether unused capacity is the main determinant of service PES.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. Existing rooms permit short-term extra sales; total rooms cap further response.

Expected diagram or calculation. P/Q axes; compare Q response to the same P change at matched scales.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes spare capacity from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Existing rooms permit short-term extra sales; total rooms cap further response.

Counterargument and conditions. Idle capacity requires complementary workers and inputs.

Application and real-world example. Hotel rooms and factories as named real markets where documented.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 25

Medium | SL/HL | 25 marks

Topic 2.6: Spare capacity

(a) [10] Explain how unused capacity and factor mobility interact in determining PES.

(b) [15] Using real-world examples, discuss whether a producer with spare machinery necessarily has elastic supply.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. Machinery and inputs must both be available to expand output.

Expected diagram or calculation. P/Q axes; compare Q response to the same P change at matched scales.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes spare capacity from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Machinery and inputs must both be available to expand output.

Counterargument and conditions. Idle capacity requires complementary workers and inputs.

Application and real-world example. Hotel rooms and factories as named real markets where documented.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 26

Easy | SL/HL | 25 marks

Topic 2.6: Storage

(a) [10] Explain how storing a good can make its marketed supply more price elastic.

(b) [15] Using real-world examples, evaluate whether storability is sufficient for a large supply response.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. Inventories can be released after price increase without immediate new production.

Expected diagram or calculation. P/Q axes; distinguish current marketed stock from new production.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes storage from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Inventories can be released after price increase without immediate new production.

Counterargument and conditions. Inventories are limited, may spoil or be held for expected future prices.

Application and real-world example. USDA oranges versus storable grain.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 27

Hard | SL/HL | 25 marks

Topic 2.6: Storage

(a) [10] Explain why perishable goods may have less elastic supply than storable goods in the very short run.

(b) [15] Using real-world examples, discuss whether perishability always makes market supply inelastic.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. Cannot carry stock readily; new deliveries and production still possible with time.

Expected diagram or calculation. P/Q axes; distinguish current marketed stock from new production.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes storage from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Cannot carry stock readily; new deliveries and production still possible with time.

Counterargument and conditions. Inventories are limited, may spoil or be held for expected future prices.

Application and real-world example. USDA oranges versus storable grain.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 28

Medium | SL/HL | 25 marks

Topic 2.6: Storage

(a) [10] Explain how the ability to accumulate inventories affects the timing of quantity supplied.

(b) [15] Using real-world examples, evaluate whether a price rise will always prompt stock release.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. Releasing now versus holding for expected future price depends on expectations.

Expected diagram or calculation. P/Q axes; distinguish current marketed stock from new production.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes storage from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Releasing now versus holding for expected future price depends on expectations.

Counterargument and conditions. Inventories are limited, may spoil or be held for expected future prices.

Application and real-world example. USDA oranges versus storable grain.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 29

Hard | SL/HL | 25 marks

Topic 2.6: Storage

(a) [10] Using appropriate diagrams, explain why two goods with equal production capacity can have different PES because of storage.

(b) [15] Using real-world examples, discuss whether inventory figures alone establish high PES.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. Storage creates a margin for current offered quantity, not necessarily future production.

Expected diagram or calculation. P/Q axes; distinguish current marketed stock from new production.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes storage from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Storage creates a margin for current offered quantity, not necessarily future production.

Counterargument and conditions. Inventories are limited, may spoil or be held for expected future prices.

Application and real-world example. USDA oranges versus storable grain.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 30

Medium | SL/HL | 25 marks

Topic 2.6: Storage

(a) [10] Explain the distinction between stored stock and new production in a firm's response to a higher price.

(b) [15] Using real-world examples, evaluate whether immediate increases in sales imply an increase in current production.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. Stock releases increase supplied quantity today even if output unchanged.

Expected diagram or calculation. P/Q axes; distinguish current marketed stock from new production.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes storage from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Stock releases increase supplied quantity today even if output unchanged.

Counterargument and conditions. Inventories are limited, may spoil or be held for expected future prices.

Application and real-world example. USDA oranges versus storable grain.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 31

Easy | SL/HL | 25 marks

Topic 2.6: Rising costs

(a) [10] Explain why rapidly increasing marginal costs can reduce PES.

(b) [15] Using real-world examples, discuss whether producers always expand strongly when prices rise.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. Additional units become costly and require larger price changes to justify output.

Expected diagram or calculation. Supply responsiveness diagram with P vertical/Q horizontal; input-price shift is not the same as rising marginal cost along S.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes rising costs from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Additional units become costly and require larger price changes to justify output.

Counterargument and conditions. Cost escalation may change with technology and input availability.

Application and real-world example. ONS producer input-price series can anchor conditions but not MC itself.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 32

Medium | SL/HL | 25 marks

Topic 2.6: Rising costs

(a) [10] Using appropriate diagrams, explain why a firm whose marginal costs rise slowly may have more price-elastic supply.

(b) [15] Using real-world examples, evaluate whether differences in cost structure alone explain different PES values.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. For same price rise, lower incremental cost growth permits larger Q expansion.

Expected diagram or calculation. Supply responsiveness diagram with P vertical/Q horizontal; input-price shift is not the same as rising marginal cost along S.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes rising costs from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. For same price rise, lower incremental cost growth permits larger Q expansion.

Counterargument and conditions. Cost escalation may change with technology and input availability.

Application and real-world example. ONS producer input-price series can anchor conditions but not MC itself.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 33

Hard | SL/HL | 25 marks

Topic 2.6: Rising costs

(a) [10] Explain how bottlenecks can cause costs to rise as a firm expands output.

(b) [15] Using real-world examples, discuss whether technological change always raises PES.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. Overtime, scarce components or congestion make extra units costly; tech may ease bottleneck.

Expected diagram or calculation. Supply responsiveness diagram with P vertical/Q horizontal; input-price shift is not the same as rising marginal cost along S.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes rising costs from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Overtime, scarce components or congestion make extra units costly; tech may ease bottleneck.

Counterargument and conditions. Cost escalation may change with technology and input availability.

Application and real-world example. ONS producer input-price series can anchor conditions but not MC itself.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 34

Medium | SL/HL | 25 marks

Topic 2.6: Rising costs

(a) [10] Explain the difference between a rise in input prices that shifts supply and increasing costs as output expands that influence PES.

(b) [15] Using real-world examples, evaluate whether an observed rise in market price reveals either effect.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. Input price shift changes S; rising marginal costs affect response along S.

Expected diagram or calculation. Supply responsiveness diagram with P vertical/Q horizontal; input-price shift is not the same as rising marginal cost along S.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes rising costs from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Input price shift changes S; rising marginal costs affect response along S.

Counterargument and conditions. Cost escalation may change with technology and input availability.

Application and real-world example. ONS producer input-price series can anchor conditions but not MC itself.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 35

Medium | SL/HL | 25 marks

Topic 2.6: Rising costs

(a) [10] Explain why two producers with similar capacity may have different PES if their expansion costs differ.

(b) [15] Using real-world examples, discuss whether spare capacity guarantees elastic supply.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. Cost of activating additional shifts and inputs can vary.

Expected diagram or calculation. Supply responsiveness diagram with P vertical/Q horizontal; input-price shift is not the same as rising marginal cost along S.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes rising costs from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Cost of activating additional shifts and inputs can vary.

Counterargument and conditions. Cost escalation may change with technology and input availability.

Application and real-world example. ONS producer input-price series can anchor conditions but not MC itself.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 36

Easy | SL/HL | 25 marks

Topic 2.6: Combined determinants

(a) [10] Explain how time and storability can jointly affect PES for a food product.

(b) [15] Using real-world examples, evaluate whether longer adjustment time makes a perishable good as responsive as a storable good.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. Time permits planting, storage permits immediate release; effects differ by horizon.

Expected diagram or calculation. Use clearly labelled P/Q curves only when relevant; do not conflate determinant of PES with shift of supply.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes combined determinants from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Time permits planting, storage permits immediate release; effects differ by horizon.

Counterargument and conditions. The strength and timing of each determinant must be compared in context.

Application and real-world example. USDA agricultural and manufacturing contrasts.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 37

Hard | SL/HL | 25 marks

Topic 2.6: Combined determinants

(a) [10] Explain how spare capacity and rapidly rising costs can pull PES in opposite directions.

(b) [15] Using real-world examples, discuss whether a factory's spare capacity guarantees a large quantity response.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. Idle machines help, but overtime and components can make expansion expensive.

Expected diagram or calculation. Use clearly labelled P/Q curves only when relevant; do not conflate determinant of PES with shift of supply.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes combined determinants from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Idle machines help, but overtime and components can make expansion expensive.

Counterargument and conditions. The strength and timing of each determinant must be compared in context.

Application and real-world example. USDA agricultural and manufacturing contrasts.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 38

Medium | SL/HL | 25 marks

Topic 2.6: Combined determinants

(a) [10] Explain why the PES of a product may differ between two regions with different factor mobility and inventory holdings.

(b) [15] Using real-world examples, evaluate whether a national PES figure adequately describes every region.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. Regional resources and inventories lead to distinct percentage responses.

Expected diagram or calculation. Use clearly labelled P/Q curves only when relevant; do not conflate determinant of PES with shift of supply.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes combined determinants from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Regional resources and inventories lead to distinct percentage responses.

Counterargument and conditions. The strength and timing of each determinant must be compared in context.

Application and real-world example. USDA agricultural and manufacturing contrasts.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 39

Hard | SL/HL | 25 marks

Topic 2.6: Combined determinants

(a) [10] Explain why a producer might have low PES in the current week but high PES over several years.

(b) [15] Using real-world examples, discuss whether firms should use one PES estimate for all planning decisions.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. Capacity, mobile factors and stocks evolve over time.

Expected diagram or calculation. Use clearly labelled P/Q curves only when relevant; do not conflate determinant of PES with shift of supply.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes combined determinants from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Capacity, mobile factors and stocks evolve over time.

Counterargument and conditions. The strength and timing of each determinant must be compared in context.

Application and real-world example. USDA agricultural and manufacturing contrasts.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 40

Medium | SL/HL | 25 marks

Topic 2.6: Combined determinants

(a) [10] Explain how two determinants of PES can offset one another in a market.

(b) [15] Using real-world examples, evaluate whether a single observable determinant is enough to predict PES.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. Storage can increase immediate response while steep expansion costs limit it.

Expected diagram or calculation. Use clearly labelled P/Q curves only when relevant; do not conflate determinant of PES with shift of supply.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes combined determinants from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Storage can increase immediate response while steep expansion costs limit it.

Counterargument and conditions. The strength and timing of each determinant must be compared in context.

Application and real-world example. USDA agricultural and manufacturing contrasts.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 41

Easy | SL/HL | 25 marks

Topic 2.6: Measurement

(a) [10] Explain why PES estimates require prices and quantities supplied measured over the same period.

(b) [15] Using real-world examples, discuss whether observed sales data can accurately measure PES.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. Mismatched periods or demand changes confuse percentage response.

Expected diagram or calculation. Show percentage formula if calculating; distinguish market observations from isolated own-price response.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes measurement from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Mismatched periods or demand changes confuse percentage response.

Counterargument and conditions. Concurrent shifts, heterogeneity and interval choice limit inference.

Application and real-world example. USDA planted acreage and prices are not a controlled PES calculation.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 42

Medium | SL/HL | 25 marks

Topic 2.6: Measurement

(a) [10] Explain why a producer's PES estimate need not equal PES for the whole market.

(b) [15] Using real-world examples, evaluate whether a large firm can represent an industry's responsiveness.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. Market aggregates heterogeneous producers, capacities and stocks.

Expected diagram or calculation. Show percentage formula if calculating; distinguish market observations from isolated own-price response.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes measurement from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Market aggregates heterogeneous producers, capacities and stocks.

Counterargument and conditions. Concurrent shifts, heterogeneity and interval choice limit inference.

Application and real-world example. USDA planted acreage and prices are not a controlled PES calculation.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 43

Hard | SL/HL | 25 marks

Topic 2.6: Measurement

(a) [10] Explain how a simultaneous change in demand can complicate estimation of PES from observed prices and sales.

(b) [15] Using real-world examples, discuss whether two price-quantity observations prove a supply response.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. Endogenous price and shifting demand mean causal response cannot be read from simple correlation.

Expected diagram or calculation. Show percentage formula if calculating; distinguish market observations from isolated own-price response.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes measurement from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Endogenous price and shifting demand mean causal response cannot be read from simple correlation.

Counterargument and conditions. Concurrent shifts, heterogeneity and interval choice limit inference.

Application and real-world example. USDA planted acreage and prices are not a controlled PES calculation.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 44

Medium | SL/HL | 25 marks

Topic 2.6: Measurement

(a) [10] Explain why the choice of initial quantity can affect a calculated percentage change in quantity supplied.

(b) [15] Using real-world examples, evaluate whether two valid estimates from different intervals must contradict one another.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. Different bases and intervals can give differing PES; state calculation method.

Expected diagram or calculation. Show percentage formula if calculating; distinguish market observations from isolated own-price response.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes measurement from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Different bases and intervals can give differing PES; state calculation method.

Counterargument and conditions. Concurrent shifts, heterogeneity and interval choice limit inference.

Application and real-world example. USDA planted acreage and prices are not a controlled PES calculation.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 45

Medium | SL/HL | 25 marks

Topic 2.6: Measurement

(a) [10] Explain why a numerical PES estimate may become outdated after a technological improvement.

(b) [15] Using real-world examples, discuss whether historical PES is a reliable guide for a new production process.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. Technology can change marginal costs, capacity and feasible adjustment.

Expected diagram or calculation. Show percentage formula if calculating; distinguish market observations from isolated own-price response.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes measurement from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Technology can change marginal costs, capacity and feasible adjustment.

Counterargument and conditions. Concurrent shifts, heterogeneity and interval choice limit inference.

Application and real-world example. USDA planted acreage and prices are not a controlled PES calculation.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 46

Easy | HL | 25 marks

Topic 2.6: HL primary comparison

(a) [10] Explain two reasons why PES for primary commodities is often lower than for manufactured products.

(b) [15] Using real-world examples, evaluate whether the comparison holds for every commodity and manufactured good.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. Growing cycles and dependence on natural conditions constrain short-run output.

Expected diagram or calculation. Use properly labelled supply curves only if necessary; compare percentages at common reference.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes hl primary comparison from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Growing cycles and dependence on natural conditions constrain short-run output.

Counterargument and conditions. Some extractive commodities are storable, while specialised manufacture can be slow.

Application and real-world example. USDA orange production and factory output contexts.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 47

Hard | HL | 25 marks

Topic 2.6: HL primary comparison

(a) [10] Explain how seasonal production can limit PES of an agricultural commodity.

(b) [15] Using real-world examples, discuss whether the price responsiveness of agriculture is always lower than manufacturing.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. Harvest cannot be moved instantly, though inventory can respond.

Expected diagram or calculation. No compulsory diagram; distinguish harvest from marketed supply.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes hl primary comparison from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Harvest cannot be moved instantly, though inventory can respond.

Counterargument and conditions. Different crops and manufactured capacity vary.

Application and real-world example. USDA Florida citrus and planting intentions.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 48

Medium | HL | 25 marks

Topic 2.6: HL primary comparison

(a) [10] Explain how storage may make a primary commodity more price responsive despite limited production flexibility.

(b) [15] Using real-world examples, evaluate whether primary products necessarily have low PES in the short run.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. Inventory release is distinct from harvesting more.

Expected diagram or calculation. No compulsory diagram; explain period and supplied quantity.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes hl primary comparison from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Inventory release is distinct from harvesting more.

Counterargument and conditions. Storage expense and existing stock determine effect.

Application and real-world example. Grain inventories versus fresh citrus.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 49

Hard | HL | 25 marks

Topic 2.6: HL primary comparison

(a) [10] Explain why specialised manufacturing can have lower PES than a storable primary commodity.

(b) [15] Using real-world examples, discuss whether the label 'manufactured' predicts PES reliably.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. Specialised capital and lead times can limit production while grain stocks are releasable.

Expected diagram or calculation. No compulsory diagram; distinguish product and horizon.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes hl primary comparison from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Specialised capital and lead times can limit production while grain stocks are releasable.

Counterargument and conditions. Other manufacturers have flexible assembly lines and different stores.

Application and real-world example. Aircraft production versus grain stock is a possible documented contrast.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 50

Medium | HL | 25 marks

Topic 2.6: HL primary comparison

(a) [10] Explain why comparing PES across primary and manufactured goods requires a common time horizon.

(b) [15] Using real-world examples, evaluate whether a broad sector average is useful to a particular producer.

Part (a) - Indicative answer and marking guidance

Essential concepts. PES is the percentage change in quantity supplied divided by the percentage change in the good's own price, ceteris paribus. A value above one is elastic and below one is inelastic; relevant determinants include time, mobility, capacity, storage and rising costs.

Required reasoning. Price response over weeks differs from years, possibly reversing rank.

Expected diagram or calculation. No compulsory diagram; use percentage-based rather than absolute comparison.

Common misconceptions. Do not confuse PES with a shift of supply, and do not use a demand percentage change in its calculation. For this question, check specifically that the student distinguishes hl primary comparison from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Price response over weeks differs from years, possibly reversing rank.

Counterargument and conditions. Product mix and local capacity also matter.

Application and real-world example. USDA acreage intentions compared with manufacturing inventories.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Sources and scope

The current IB Economics guide controls syllabus boundaries. The uploaded Economics for the IB Diploma, second edition (2012), is a supplementary conceptual check; its older chapter sequence and content cannot expand the current specification. Real-world examples are optional and should be checked against their cited sources.

G. IB Economics guide, first assessment 2022, Topic 2.6, printed p. 28.

Open source

A. USDA ERS, Florida orange production affected by disasters and disease, 2024.

Open source

B. USDA ERS, changing planted acreage and crop-price incentives, 2024.

Open source

Final checks

50 paired questions; 10 Easy / 25 Medium / 15 Hard; Questions 1-45 are SL/HL. Questions 46-50 assess the HL-only comparison of primary commodities and manufactured products. The student paper contains no answers or difficulty labels.